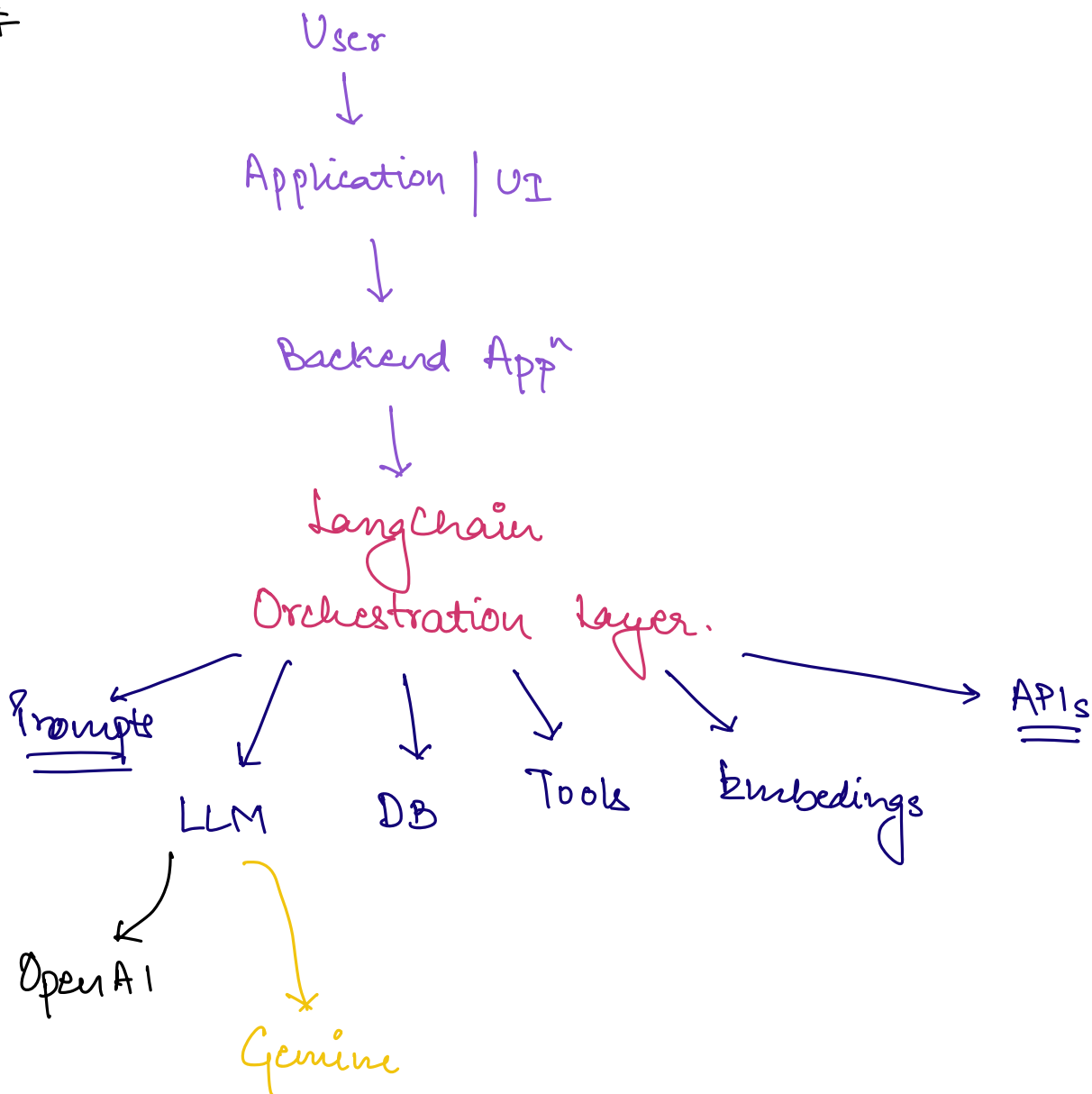


## # langChain

→ It is an open-source framework for building LLM powered application by connecting models, prompts, tools, memory, retrievers & workflows in a structured way.

#



LANGCHAIN

In langchain, we build applications by connecting small reusable components together.

These components are called as Runnables.

A Runnable is a unit of work that can be invoked with other components.

Examples.

- LLM
- Retrievers
- Tools
- Chain
- OutputParser.
- PromptTemplate

Input → PromptTemplate → LLM → OutputParser  
↓  
Response.

## Chain

→ Sequence of Connected Steps.

## Agent

→ Agent can decide the next step

## # Core Components of LangChain

→ Models

→ Prompts

→ Chains

→ Memory

→ Indexes

→ Agents

## # Models.

→ LLMs → <sup>Generation</sup> Reasoning

→ Chat Models

→ Text Embedding Models.

## # Prompts

→ Instructions we send to the model to get the desired output.

→ Prompt Templates.

↳ Template blueprints which uses placeholders/variables to generate dynamic inputs.

→ Chat Prompt Templates.

↳ For chat models, langChain offers template to form list of messages.

→ Example Selectors

↳ To improve the prompt performance, we can include some examples (few-shot prompting).

## # Chains.

→ Sequence of connected steps that processes input through multiple components.

→ Pipeline.

→ LLM Chain.

↳ Combines prompt template, model & output parsers (optional).

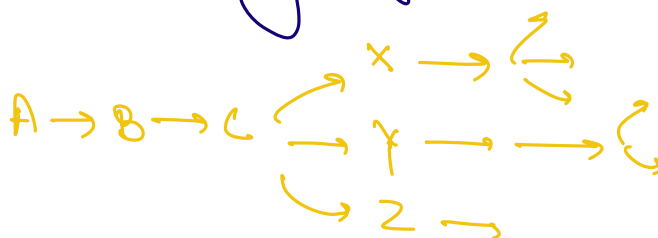
→ Sequential & Composite Chains:



$A \rightarrow B \rightarrow C \rightarrow D \rightarrow E$



Branching Logic



→ Parallel Chaining

# Memory

↳ Helps to maintain the context within Conversations.

⇒ Integrating memory with chains or agents enables them to generate context-aware & coherent responses over multiple turns.

## # Indexes.

→ Indexes are used to organise & retrieve document data, which is essential for RAG based applications.

## # Agents.

Agents are dynamic, decision-making components that can select actions or tools based on the user input.

⇒ Prompt Template.

↳ Reusable prompt structure with variables.

Prompt: "Explain SQL Indexes to beginner students in simple words"



template: "Explain {topic} to {audience} students in {style} words"

↓  
Reusability